

Prateek

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EDUCATION

Indian Institute of Management Mumbai

2026 – 2028

Master of Business Administration (Core)

Mumbai, India

Indian Institute of Information Technology Guwahati

2021 – 2025

Bachelor of Technology, Computer Science and Engineering

Guwahati, India

TECHNICAL COMPETENCIES

Languages: C++, C#, Java, Python, JavaScript, TypeScript, SQL

Data Engineering & Cloud: Microsoft Azure, Microsoft Fabric, Azure Synapse Analytics, Azure Data Factory, Azure Data Lake Storage, PySpark, Kusto Query Language (KQL), SQL, Azure DevOps, CI/CD, ETL Pipelines, Data Warehousing

Full Stack: Next.js, React, React Native, Node.js, Flask, .NET, HTML, CSS

Developer Tools: Git, GitHub, Docker, VS Code, JIRA, Unity, AntiGravity

EXPERIENCE

MAQ Software

Jan 2025 – May 2026

Associate Software Engineer → Software Engineer 1

Noida, India

- Built and owned production data engineering pipelines for Microsoft Azure Core Business Insights, enabling business-critical operational reporting and executive decision-making on global Azure VM telemetry.
- Owned 4–5 production ETL processes built on Azure Synapse, Microsoft Fabric, PySpark, SQL and KQL, supporting business insights generated from hundreds of millions of telemetry records processed daily.
- Optimized long-running production workflows, reducing execution time from approximately 12 hours to 3 hours through query optimization, pipeline redesign and workflow improvements, significantly improving reporting freshness.
- Reduced recurring monitoring incidents from approximately 10–11 per day to 3–4 per day by improving validation logic, production reliability and monitoring workflows across critical data pipelines.
- Contributed to modernization of a metadata-driven ingestion architecture replacing 300+ legacy pipelines and 400+ tables with parameterized frameworks, substantially improving maintainability and reducing operational complexity.
- Developed ingestion and transformation frameworks spanning ADLS, SQL, KQL, Blob Storage and Lakehouse environments, supporting 50+ production processes and 60+ business intelligence reports.
- Collaborated closely with Microsoft stakeholders, monitoring teams, S360 engineers, developers and product teams to prioritize incidents, resolve production issues and deliver data-driven business insights.

PROJECTS

Hybrid Memory Aware Cache Replacement Policy Design [GitHub](#) — [Deck](#)

Nov 2024

- Designed and implemented hybrid memory-aware cache replacement policies over baseline LRU for DRAM/NVM architectures using the ChampSim simulator under the guidance of Dr. Arijit Nath.
- Developed frequency-aware and channel-aware eviction strategies (P1 and P2) by incorporating memory access locality, recency and channel prioritization into victim selection algorithms.
- Evaluated policies across benchmark traces on a 4-channel hybrid memory architecture, reducing writebacks while maintaining IPC and achieving up to 5% improvement in LLC hit rate under selected workloads.

Research Literature Recommendation & Academic Domain Prediction [GitHub](#) — [Deck](#)

Nov 2024

- Built an NLP-powered research discovery platform combining semantic paper recommendation with multi-label academic domain classification to accelerate academic literature discovery.
- Developed a TensorFlow-based classifier leveraging TF-IDF vectorization and deep neural networks, achieving 99.4% categorical accuracy on validation and test datasets.

- Implemented semantic retrieval using all-MiniLM-L6-v2 sentence-transformer embeddings and cosine similarity to recommend contextually similar research papers.

Interactive 3D Developer Portfolio [Live](#) 2026

- Built an interactive 3D portfolio in Next.js and TypeScript with optimized rendering and reusable component architecture.
- Implemented performant animation pipelines and responsive UI patterns to deliver production-grade user experience.

YouTube Advertisement Recommendation System [GitHub](#) May 2024

- Built a machine learning pipeline to recommend advertisements by classifying YouTube videos using textual metadata from 3,500+ videos and 6,600+ advertisement records.
- Engineered TF-IDF unigram and bigram features through preprocessing, lemmatization and feature extraction before training Naive Bayes and SVM classifiers.
- Achieved 97% and 98% classification accuracy respectively while evaluating performance through precision-recall analysis and confusion matrices.

LEADERSHIP

Yuvaan Cultural Fest, IIIT Guwahati | PR Team Member → Head 2023 – 2024

- Led a PR team of 7-8 members and coordinated outreach with 8+ regional colleges including Gauhati University, RGU and Don Bosco University to drive participation and engagement.
- Spearheaded artist onboarding and external partnerships, coordinating with Bharat Bass Festival and facilitating performances by Zephyrtone, DJ Elina Chauhan and VJ Infinity.
- Coordinated inter-college communication, stakeholder management and event logistics while leading a team of 10 volunteers to ensure seamless execution of festival activities.

ACHIEVEMENTS & CERTIFICATIONS

NTSE Scholar	2019
Microsoft Certified: DP-700	2025
Microsoft Certified: DP-600	2025

INTERESTS

Product Management · Supply Chain Analytics · Operations · AI Systems · Data Engineering · Distributed Systems · LLM Applications